



THERMOBIO ENERGY & ENGINEERING SOLUTIONS

INTEGRATED INDUSTRIAL ENGINEERING SERVICES

Operations & Maintenance • Power Plant Operations • Electrical & Instrumentation • PLC / DCS / SCADA • Industrial Automation

SUSTAINABLE ENERGY
RELIABLE OPERATIONS
ENGINEERING EXCELLENCE

Ethanol & ENA Plants • Power Plants • Electrical & Instrumentation • Automation & Digital Engineering

Company Profile

ThermoBio Energy & Engineering Solutions is an integrated industrial engineering service partner focused on plant operations, maintenance, commissioning and automation. Our service approach combines operations knowledge with electrical, instrumentation, control and automation capabilities so that project and plant teams can work with a coordinated engineering partner.

We support industrial energy and process environments where dependable operation, disciplined maintenance, commissioning quality and practical troubleshooting are important. Our capabilities cover ethanol and ENA plants, power-generation systems, industrial electrical systems, field instrumentation and PLC/DCS/SCADA-based automation.

Vision	To contribute to safer, more reliable and more efficient industrial energy and process operations through practical engineering and technology.
Mission	Provide dependable O&M;, commissioning, E&I; and automation support with a focus on operational continuity, engineering discipline, maintainability and measurable improvement.
Approach	Understand the plant, identify the operational requirement, execute systematically, document the work and support the team through stabilization and improvement.

Key Industrial Focus

Ethanol & ENA Plants	Power / Thermal Plants
Electrical Systems	Instrumentation & Control
PLC / DCS / SCADA	Industrial Automation

Leadership & Technical Expertise

Experienced professionals supporting plant operations, commissioning, electrical & instrumentation and industrial automation.

Rahul Kumar Yadav Director 10+ years expertise Ethanol & ENA Plant Operations • Ethanol and ENA plant operations and maintenance • Plant commissioning, start-up and shutdown coordination • Process, utility and equipment performance monitoring • Preventive maintenance, troubleshooting and reliability improvement • Production coordination, SOP implementation and technical reporting • Safe and efficient plant operation through cross-functional coordination	Guddu Kumar Director 8+ years expertise Electrical & Instrumentation • Electrical and instrumentation engineering for power plants • Installation, testing, commissioning and maintenance • MCC, PCC, switchgear, motors, transformers and VFD systems • Field instrumentation, calibration and loop checking • PLC/DCS/SCADA interface and control-system troubleshooting • Breakdown analysis, preventive maintenance and plant reliability
Aman Kumar Singh Director 6+ years expertise Power Plant Operations & E&I; • Power plant operation and maintenance • Electrical and instrumentation commissioning support • Boiler, turbine, generator and auxiliary-system coordination • Start-up, shutdown, parameter monitoring and troubleshooting • Preventive and breakdown maintenance of E&I; systems • Safety-focused operations and cross-functional plant coordination	Rahul Kumar Director 5 years expertise E&I; • Automation • R&D; • Electrical and instrumentation engineering for industrial plants • PLC, DCS, SCADA and industrial automation • Control logic, interlocks, sequences and instrumentation integration • Commissioning, troubleshooting and control-loop optimization • Engineering R&D;,, digital plant monitoring and data-driven solutions • Practical automation and predictive-maintenance concept development

Our Services — Detailed Capabilities

Flexible support for long-term O&M, commissioning projects, maintenance assignments and industrial automation requirements.

Ethanol Plant O&M; Integrated support for ethanol and ENA plant operations, utilities and maintenance. • Process and utility operations • Preventive and breakdown maintenance • Shift support, reporting and troubleshooting • Equipment performance and reliability improvement	Power Plant Operations Operational and maintenance support for boilers, turbines, generators and balance-of-plant systems. • Start-up, shutdown and normal operation support • Operating parameter monitoring • Boiler, turbine and generator coordination • Electrical, instrumentation and control coordination
Electrical Services Industrial electrical commissioning, testing, maintenance and troubleshooting. • MCC/PCC and switchgear • Motors, VFDs and transformers • Testing, fault finding and maintenance • Electrical equipment commissioning support	Instrumentation & Control Field instrumentation and control-system support for reliable measurement and process control. • Flow, pressure, temperature and level instruments • Calibration and loop checking • Control valves and signal verification • I/O checks and troubleshooting
PLC / DCS / SCADA Configuration, programming and troubleshooting support for industrial control systems. • PLC logic and modifications • DCS configuration and control strategies • HMI/SCADA support and tag integration • Sequences, permissives, interlocks and trips	Industrial Automation Automation engineering focused on safe sequences, control performance and plant optimization. • Sequence and interlock development • PID/control-loop support • VFD and equipment integration • Control optimization and digital monitoring concepts

Commissioning & Project Support

A structured execution approach from readiness checks through start-up, stabilization and handover.

01	Pre-Commissioning & Readiness Systematic readiness checks before energization or start-up: installation verification, equipment status, cable checks, instrument availability, documentation review and punch-point identification.
02	Electrical Commissioning Functional verification of panels, MCC/PCC, switchgear, motors, VFDs and transformers, including control supply checks, interlock verification and start/stop testing.
03	Instrumentation & Control Calibration support, loop checking, I/O verification, signal validation, control-valve checks and field-to-control-room confirmation for reliable measurement and control.
04	PLC / DCS / HMI I/O mapping, logic verification, sequence testing, permissives, interlocks, alarms, trips and HMI/SCADA indication checks before controlled operation.
05	Start-Up & Stabilization Controlled energization and equipment start-up with operating-parameter monitoring, troubleshooting, tuning support and coordination between process, electrical and instrumentation teams.
06	Handover & Project Support Punch-point closure, test records, commissioning documentation, operating support and technical coordination during the transition from project execution to stable operation.

Project Support Scope

Technical coordination, site readiness, testing records, punch-point management, troubleshooting during start-up, operating documentation and support during stabilization and handover.

Digital Engineering & Technology

Turbine Diagnostic Software

A digital inspection and diagnostic workspace for turbine operating parameters, alarms, performance indicators, history and engineering review.

Power System Tool

A browser-based power-system analysis workspace covering load-flow, short-circuit, protection/TCC, transient-stability, single-line and related engineering study functions.

Industrial Automation & R&D;

PLC/DCS/SCADA engineering, control optimization, data-driven monitoring and practical R&D; concepts for improving plant visibility and maintenance decisions.

Digital tools are intended to support engineering workflows; project-specific calculations and safety-critical decisions should be validated against applicable standards, OEM information and site data.

Contact ThermoBio

Let's discuss your O&M, commissioning, E&I, automation or industrial engineering requirement.

Company	ThermoBio Energy & Engineering Solutions
Email	tbenergy.engg@gmail.com
Website	https://tbenergyengg.github.io/thermobio-energy/
Office 1	Office 1: Deoria, Uttar Pradesh - 274001, India
Office 2	Office 2: Hajipur, Bihar - 844101, India
Mobile	+91 89488 07015 +91 95722 42477 +91 90067 44367 +91 91986 67587



Scan to open the ThermoBio website

SUSTAINABLE ENERGY | RELIABLE OPERATIONS | ENGINEERING EXCELLENCE

This catalogue describes the company's service capabilities and engineering focus. Detailed project scope, deliverables and commercial terms can be defined for each assignment.